



Full length article

Fetal agenesis of the corpus callosum: Clinical and genetic analysis in a series of 40 patients

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ARTICLE INFO

Keywords:

Prenatal diagnosis
Agenesis of the corpus callosum
Whole-exome sequencing
Copy number variation sequencing

ABSTRACT

Objectives: This study aimed to explore the genetic causes of agenesis of the corpus callosum (ACC) and assess the utility of karyotype analysis, copy number variation sequencing (CNV-seq), and whole-exome sequencing (WES) to genetically diagnose fetal ACC.

Methods: We retrospectively examined 40 fetuses diagnosed with ACC who underwent prenatal ultrasonography or magnetic resonance imaging between January 2019 and October 2023. Genetic tests were conducted on the fetuses using karyotype analysis or CNV-seq as the first-line diagnosis. WES was performed if aneuploid and pathogenic CNVs were excluded.

Results: Among the 40 fetuses, 29 (72 %) had non-isolated ACC and 11 (28 %) had isolated ACC. Cerebellar dysplasia and hydrocephalus were the most common abnormal developments in the central nervous system. Twenty-eight patients underwent karyotype analysis, with a detection rate of 14 % (4/28). Twenty-six patients underwent CNV-seq; three patients were found to have pathogenic CNVs, with a detection rate of 12 % (3/26). Thirty-three fetuses with no findings of karyotype analysis or CNV-seq were subsequently tested using WES, with a detection rate of 36 % (12/33). Overall, the total diagnostic yield was 48 % (19/40), and monogenic etiology accounted for 30 % (12/40). The genetic detection rate of fetal non-isolated ACC (62 %, 18/29) was higher than that of isolated ACC (9 %, 1/11).

Conclusion: Prenatal genetic analysis of fetuses with ACC is clinically significant, with monogenic disorders being the main cause. WES may enhance the detection rate of fetuses with ACC with negative karyotype analysis or CNV-seq results.

Introduction

Corpus callosum (CC), comprising approximately 190 million axons, is the main telencephalic commissure of the human brain and is involved in interhemispheric communication [1]. The incidence of agenesis of the corpus callosum (ACC) in newborns is estimated to be 3–7 per 1000 and 2–3 % in those with impaired neurological development [2,3]. According to morphological abnormalities, CC can be divided into complete ACC (cACC), partial ACC, or dysgenesis of CC. ACC is usually associated with other cerebral and extracerebral anomalies and is a feature of >200 syndromes [4–6]. The etiology of CC anomalies resulting from chromosomal, monogenic, infectious, vascular, toxic, or teratogenic causes remains unknown in most cases [7]. Clinical sequelae range from mild or no manifestations to severe neurodevelopmental

disabilities. Recent meta-analyses have shown that in the absence of this associated abnormality, isolated ACC leads to normal long-term developmental outcomes in approximately 60–70 % of infants [8].

Genetic etiology studies of fetal ACC are often combined with karyotype analysis and genome sequencing (copy number variation sequencing [CNV-seq]). Conventional karyotype analysis can detect fetal aneuploidy, chromosomal balanced translocations, and unbalanced structural rearrangements >5 Mb. Trisomy 18, 13, and 21, mosaic trisomy 8, and monosomy X are frequently detected using karyotype analysis, and these chromosomal abnormalities have been associated with fetal non-isolated ACC [9,10]. However, fetal ACC is associated with some crucial genetic syndromes that cannot be detected by karyotype analysis. In the genetic analysis of prenatal ultrasound (US) indicating fetal structural abnormalities, the detection rate of genome

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Received 13 December 2023; Received in revised form 1 May 2024; Accepted 7 May 2024

Available online 8 May 2024

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sequencing is 6–10 % higher than that of karyotype analysis [11,12]. Therefore, it has recently become the recommended primary genomic test in cases of fetal structural abnormalities. Special regions, including 1q36, 1q42-43, 4p16.3, and 6qter, exhibit changes in these fetuses [7,10,13]. However, CNV-seq can only detect genome copy number variations (CNVs) > 100 kb in specific targeted regions.

Whole-exome sequencing (WES) has received more attention for allowing the simultaneous detection of nucleotide sequences of most expressed genes in the genome and for its lower cost, smaller amount of fetal DNA required, relatively faster turnaround possibilities, and greater sequencing depth. This technique has recently been introduced to prenatal diagnosis and is estimated to have a high diagnostic rate for congenital malformed fetuses [14]. According to the postnatal literature, WES has an estimated diagnosis rate of 31 % for brain abnormalities [15].

This study aimed to investigate the application value of chromosome karyotype analysis, CNV-seq, and WES techniques in the prenatal diagnosis of fetuses with ACC at the cytogenetic, genomic, and single-gene levels.

Methods

Patient recruitment

This study included 40 fetuses diagnosed with ACC with or without other anomalies (central nervous system (CNS) or multisystem) using magnetic resonance imaging or prenatal ultrasound at the First Affiliated Hospital of Zhengzhou University between January 2019 and October 2023. MRI and ultrasound reports provide information on fetal phenotypes. We included those with prenatal imaging confirmation of isolated or non-isolated ACC in the fetus and had undergone WES following negative CNV-seq/karyotyping. Those who did not undergo WES following negative CNV-seq/karyotyping and pregnant women with a history of teratogens, infections, diabetes, hypertension, or other common diseases were excluded. We collected amniotic cells and peripheral blood samples from parents. The present study was approved by the ethics committee (KS-2018-KY-36) of the First Affiliated Hospital of Zhengzhou University, and informed consent was obtained from all the parents of the fetuses.

DNA extraction and quantitative fluorescent (QF)-PCR

For pregnant women between 19 and 28 weeks of gestation, amniocentesis was performed under US guidance to extract 10 to 20 mL of amniotic fluid. The DNA extraction kit (Omega Bio-Tek, USA) was used to extract DNA from the amniotic fluid. We drew 2 mL of peripheral venous blood from pregnant women and their husbands using EDTA-K2 tubes. The genomic DNA of parents were extracted using the QIAamp DNA extraction kit (Qiagen, Germany). QF-PCR was performed as previously described [16] and was used as quality control to detect the presence of maternal DNA contamination in fetal samples.

Karyotype analysis

For culture, approximately 20 mL of amniotic fluid was used. The standard protocol was followed when performing conventional karyotyping analysis with cultured cells using G-banding at 320–400 band resolution. They were cultivated at 37 °C with 5 % CO₂. Colchicine was added to the harvest cells before harvest, followed by the addition of potassium chloride hypotonic solution. G-banding was a regular procedure. Five karyotypes were examined, and 20 mitotic phases were counted. The ISCN 2020 was followed in describing the karyotypes.

Copy number variation sequencing

The CNV-seq protocol was followed as previously described [17].

The CNVs were mapped to the GRCh37/hg19 reference genome after identification. We used databases of Genomic Variants (<https://dgv.tcag.ca/dgv/app/home>), Database of Chromosomal Imbalance and Phenotype in Humans Using Ensembl Resources (<https://decipher.sanger.ac.uk/>), ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>), Online Mendelian Inheritance in Man (OMIM) database (<https://omim.org/>), and ClinGen (<https://www.clinicalgenome.org/>) to interpret and categorize the clinical significance of candidate CNVs following previously published guidelines.

Whole-exome sequencing

For WES, the parents and fetus trios approach was employed. In short, exomes were sequenced using standard procedures and DNA extracted from either blood or amniotic fluid. Qubit 4.0 (Thermo Fisher Scientific Inc. USA) was used for quantification. Exome coding and splicing regions were captured using the AIExome Human Exome Panel V2 Plus with TargetSeq One Hyb & Wash Kit (iGeneTech Co., Ltd, Beijing, China), following the manufacturer's instructions. DNA libraries were established using the Ada & Index Kit (UDI for LIM) and Enzyme Plus Library Prep Kit (iGeneTech Co., Ltd, Beijing, China). The acquired libraries were then sequenced with an average sequencing depth >100X and 20X sequencing coverage >98 % using the Illumina NovaSeq6000 platform (San Diego, CA, USA). Sentieon (release 201,808.05) was utilized to align paired-end reads with the human reference genome GRCh37/hg19 following the best practices recommended by the Genome Analysis Toolkit. Picard version 2.9.0 was then employed to eliminate duplicate reads. Potential variations were performed following the American College of Medical Genetics and Genomics guidelines [18]. All the chosen variants were categorized as pathogenic (P), likely pathogenic (LP), variants of uncertain significance (VUS), likely benign (LB), or benign (B).

Molecular model

Using the homology-modeling server SWISS-MODEL (<https://swissmodel.expasy.org/>), a structural model of the novel missense variants in *TUBB2B* and *TUBA1A* were created. Using PyMOL protein model structure simulation software, the target protein sequence's three-dimensional structure was built in order to assess the impact of amino acid substitution on protein.

Follow-up of pregnancy outcomes

Following a precise prenatal diagnosis, we offered each qualified pregnant woman comprehensive genetic counseling that covers the prognosis, treatments, and clinical signs of the disease. The decision to terminate the pregnancy was then up to the expectant mother and her family. Parents of all fetuses were monitored via telephone calls and electronic medical record system to assess pregnancy outcomes, diagnoses, and postpartum care.

Results

Case characteristics

We retrospectively included 40 fetuses with ACC between January 2019 and October 2023. The mean maternal age was 27 years, and the mean gestational age was 24 weeks. Four pregnant women had a history of pregnancy with similar congenital malformations. The fetuses were 50 % (n = 20) females and 50 % (n = 20) males. Among the 40 patients, 24 (60 %) fetuses had cACC, 8 (20 %) had pACC, and 8 (20 %) had CC dysgenesis. At the time of inclusion, 11 (28 %) fetuses had isolated ACC, and 29 (72 %) fetuses had non-isolated ACC. In non-isolated cases, other brain abnormalities include hydrocephalus (n = 4), choroid cyst (n = 2), posterior fossa abnormalities (n = 2), widening of longitudinal fissure (n

= 2), vermian hypoplasia (n = 3), cerebellar dysplasia (n = 4), microcephaly (n = 1), periventricular white matter injury (n = 1), ependymal cyst (n = 1). Deformities outside the brain include heart deformities (n = 4), abnormal posture of both hands (n = 1), clubfeet (n = 2), fetal growth restriction (n = 2), micrognathia (n = 1), polyhydramnios (n = 2), no gastric vesicles (n = 1), esophageal atresia (n = 1), absence of radius (n = 1), and diaphragmatic hernia (n = 1). Detailed information is provided in Table 1 and the research flow chart is shown in Fig. 1.

Genetic analysis

Among the 40 fetuses with ACC, 28 were selected for karyotype analysis, and the detection rate was 14 % (4/28). Twenty-six were selected for CNV-seq, and the detection rate of pathogenic CNV was 12 % (3/26). Detailed information is provided in Table 2. Thirty-three fetuses with no findings of karyotype analysis or CNV-seq were subsequently tested using WES. Twelve fetuses were diagnosed with monogenic disease, with a diagnosis rate of 36 % (12/33). Of the 12 diagnosed cases, 50 % (5/12) were associated with autosomal dominant disease (AD), 17 % (2/12) with autosomal recessive disease (AR), and 33 % (4/12) with X-linked recessive disease. A total of 9 genes and 14P/LP variants were detected in 12 cases of single-gene variants, including *L1CAM* (4/14), *TUBA1A* (1/14), *TUBB2B* (1/14), *SON* (1/14), *RAB3GAP1* (2/14), *SPTAN1* (1/14), *AMPD2* (2/14), *ZEB2* (1/14), and *ZBTB20* (1/14). Of these, 4 were novel variants, and 2 were identified: *TUBA1A* gene c.83A > T (p. His28Leu) and *ZEB2* gene c.2481_2488dup-CAAAAGTA (p.Ile830fs), inherited from the mother, who showed no clinical manifestations. Detailed information is provided in Tables 1 and 3. Among 40 fetal ACC cases, 11 were isolated ACC, of which 1 (9 %) case had chromosome abnormality. Twenty-nine cases were non-isolated ACC, of which 6 (21 %) had chromosomal abnormalities and 12 (41 %) had single-gene mutations. The overall genetic diagnosis rate in the 40 fetuses with ACC studied was 48 % (19/40). The genetic detection rate of non-isolated ACC (62 %, 18/29) was much higher than that of isolated ACC (9 %, 2/11).

Structural analysis for novel missense variants

The SWISS-MODEL was used to construct the tertiary structure of proteins, and the PyMol was applied to predict protein amino acid interactions. In *TUBB2B*, Ser275 was located in the highly conserved β -tubulin intermediate domain [19], and Ser275 was located in the random coil of the secondary structure, forming hydrogen bonds with Ser278 and Gln279. After variation to asparagine, it formed hydrogen bonds with Ser278, Gln279, and Gly277. This causes changes in the number of hydrogen bonds, affects the binding of GTP/GDP, hinders the formation of tubulin heterodimers, changes the stability of microtubules, and causes neurological diseases [20]. In *TUBA1A*, His28 was located at the guanine nucleotide-binding region [19], and the secondary α -helical structure of His28 forms A hydrogen bond with Tyr24 with a bond length of 3.2 Å. After variation to Leucine, the hydrogen bond was slightly lengthened to 3.3 Å. The change of hydrogen bond length may have caused the abnormality of guanine nucleotide-binding region and affected the function of tubulin [20].

Pregnancy outcomes and follow-up

Among the 40 patients, 5 couples (13 %) continued with the pregnancy and 35 couples (87 %) opted to terminate the pregnancy. Of the 19 cases without a genetic cause, 5 (26 %) were born alive. Telephone follow-up revealed one patient had umbilical hernia and occult spina bifida at birth, one had growth retardation at 3 years and 5 months, and the remaining 3 cases were not abnormal at 11 months, 13 months, and 3 years.

Discussion

This study performed a genetic analysis for 40 fetal ACC cases with an overall detection rate of 48 % (19/40), which was slightly higher than that of 44 % (44/138) and 32 % (13/41) reported in previous ACC studies [21,22]. Sample size and inclusion criteria can account for variations in detection rates. Our study included limited cases, including isolated (28 %, 11/40) and non-isolated ACC (72 %, 29/40). The genetic detection rate of non-isolated ACC (62 %, 18/29) was much higher than that of isolated ACC (9 %, 2/11). This indicates poor prognosis of non-isolated ACC and is consistent with the higher rate of chromosomal abnormalities found in fetal non-isolated ACC [23]. When fetal karyotype analysis or CNV-seq was used as the first-line diagnosis, the detection rates were 14 % and 12 %, respectively. If aneuploidy and pathogenic CNVs were excluded, WES was performed on fetal ACC, and the detection rate of single-gene variants was 36 %. WES provided an additional genetic diagnosis in approximately 30 % of cases compared with karyotype analysis and CNV-seq alone.

Current standard prenatal procedures are limited to investigating chromosomal abnormalities through fetal karyotype analysis, chromosomal microarray analysis (CMA) or CNV-seq. The detection rate of karyotype analysis in this study was 14 % (4/28). The most common form of chromosome number abnormality reported in ACC cases is trisomy 18[24]. In this study, 2 (5 %) cases of trisomy 18 were detected, which confirmed the above conclusion. High-throughput sequencing methods, including CNV-seq, are increasingly being used in prenatal diagnosis in China. The technique has been used frequently as a substitute for traditional cytogenetics, particularly when invasive testing is recommended. In this study, the detection rate of CNV-seq was 12 % (3/26), which confirmed that pathogenic CNVs may be a genetic cause of fetal ACC. Overall, the diagnosis rate of chromosomal abnormalities in this study was 18 % (7/40), which was basically consistent with the diagnosis rate in the two large cohorts of ACC (16 %, 17 %)[13,25]. However, most ACC cases still have no known cause.

Many genes involved in the CC process developed to encode microtubules or metabolic enzyme components, including *SATB2*, *FOXP1*, *GLI3*, *KIF7*, and *L1CAM* genes, have been reported [26–28]. WES has been recommended in many studies to identify genetic causes. A meta-analysis [29] showed that WES provided up to 43 % additional genetic diagnosis compared with karyotyping and CMA alone. In this study, single-gene variants undetectable by karyotype analysis and CNV-seq were detected in 12 fetuses, with an overall diagnosis rate of 30 %. We also classified 14 genetic variants in nine genes associated with prenatal ACC and provided a complete description of prenatal phenotypes. This helps elucidate the potential genetic etiology of fetal ACC and establish the causal relationship of the fetal phenotype to make the disease diagnosis more accurate. Mustafa et al. [30] reported a detection rate of approximately 55 % for WES of non-isolated ACC and 32 % for isolated ACC. The WES detection rate of fetal non-isolated ACC (62 %, 18/29) was much higher than that of isolated ACC (9 %, 1/11) in this study. This suggests greater genetic risks of fetal non-isolated ACC. The aqueduct of Sylvius (HSAS; OMIM:307000) was the most frequently diagnosed condition in this study, which was caused by *L1CAM* in 4 cases. The four cases were all male, and three of them shared a family history of unfavorable pregnancies. The novel *TUBA1A* and *ZEB2* gene variants associated with ACC were found in two cases. The mutation originated in the mother but was not clinically present, which may be due to the incomplete penetrance or gonadal mosaicism. Using WES to identify the genetic causes of fetal imaging abnormal phenotypes can help select prenatal WES in subsequent pregnancies to determine if the fetus is affected by the same disease.

Owing to ACC rarity, counseling affected parents remains controversial and difficult. Comprehending the genetic heterogeneity of ACC could be crucial for identifying the condition, creating early intervention plans, and helping parents make educated family planning decisions. We discovered that WES is feasible and in most cases, results can be

Table 1
Clinical characteristics of the participants in this study.

NO.	Maternal age	G (weeks)	Sex	Type of ACC	Prenatal US /MRI	Method	Outcome
1	19	22 ⁺¹	F	cACCIsoIated	Abnormal shape of CSP, cACC	CNV-seq, WES	TOP
2	29	26	M	pACCIsoIated	Absent CSP, pACC	Karyotype analysis, CNV-seq, WES	TOP
3	24	23 ⁺³	M	pACCIsoIated	Absent CSP, pACC	Karyotype analysis, WES	TOP
4	25	22	M	pACCIsoIated	Absent CSP, pACC	Karyotype analysis, CNV-seq, WES	LB
5	28	27	M	Dysgenesis CCIsoIated	The splenium dysplasia of the CC	CNV-seq, WES	LB
6	28	25 ⁺³	F	Dysgenesis CCIsoIated	Abnormal shape of CSP,Dysgenesis CC	Karyotype analysis, CNV-seq, WES	TOP
7	26	24 ⁺²	F	cACCIsoIated	dilated lateral ventricles, cACC	CNV-seq, WES	TOP
8	24	27 ⁺²	F	Dysgenesis CCIsoIated	Abnormal shape of CSP, Dysgenesis CC	CNV-seq, WES	LB (developmental retardation)
9	24	24	F	Dysgenesis CCIsoIated	Dysgenesis CC, dilated lateral ventricles	Karyotype analysis, CNV-seq, WES	LB (birth is accompanied by umbilical hernia and occult spina bifida)
10	37	23 ⁺³	F	cACCIsoIated	Absent CSP, dilated lateral ventriclescACC	Karyotype analysis,CNV-seq	TOP
11	26	27 ⁺⁵	M	Dysgenesis CCIsoIated	Abnormal shape of CSP, Dysgenesis CC	Karyotype analysis,CNV-seq,WES	LB
12	25	23 ⁺⁴	F	cACCNon-isolated	Hydrocephalus, cACC	CNV-seq, WES	TOP
13	29	24 ⁺¹	F	cACCNon-isolated	widening of the longitudinal fissure, vermian hypoplasia, cACCabnormal posture of both hands	Karyotype analysis, WES	TOP
14	19	21 ⁺¹	F	cACCNon-isolated	Hydrocephalus, dilated lateral ventricles, cACC	Karyotype analysis,WES	TOP
15	26	25	F	cACCNon-isolated	Hydrocephalus, Absent CSP, cACC	CNV-seq, WES	TOP
16	31	28	M	pACC,Non-isolated	Absent CSP, pACC, dilated lateral ventricles, posterior fossa abnormalities	Karyotype analysis, WES	TOP
17	31	24 ⁺³	F	Dysgenesis CCNon-Isolated	Abnormal shape of CSP, abnormal cerebral falx, dysgenesis CC	Karyotype analysis,WES	TOP
18	20	21 ⁺¹	M	pACCNon-isolated	Abnormal shape of CSP, pACCCTOF	CNV-seq, WES	TOP
19	25	22 ⁺⁴	F	Dysgenesis CC Non-isolated	Dysgenesis CCclubfeet	Karyotype analysis,WES	TOP
20	27	23 ⁺²	F	cACCNon-isolated	Absent CSP, cACCFCGR	Karyotype analysis,WES	TOP
21	23	22	F	cACCNon-isolated	Microcephaly, Absent CSP, dilated lateral ventricles	Karyotype analysis, CNV-seq, WES	TOP
22	32	25	M	cACCNon-isolated	upward displacement of the third ventricle, cACC Absent CSP, dilated lateral ventricles, upward displacement of the third ventricle, The longitudinal cisterna of the brain widened, left choroid plexus cys	Karyotype analysis, CNV-seq, WES	TOP
23	28	23 ⁺⁶	M	pACCNon-isolated	Left choroid plexus cys, EIF,pACC	Karyotype analysis, CNV-seq, WES	TOP
24	26	23	M	cACCNon-isolated	Cerebellar hypoplasia, absent CSP, cACC	CNV-seq	TOP
25	30	21 ⁺³	F	cACCNon-isolated	Polyhydramnios, VSD, esophageal atresia, abnormal posture of both hands, absence of right radius, cACC	Karyotype analysis	TOP
26	31	26 ⁺³	M	pACCNon-isolated	Polyhydramnios, VSD, SUA, absent nasalbone, overlapping finger, diaphragmatic hernia, ependymal cyst, pACC	Karyotype analysis	TOP
27	31	22 ⁺¹	M	pACCNon-isolated	Extensive periventricular white matter injury, dilated lateral ventricles, VSD, pACC	Karyotype analysis	TOP
28	22	25 ⁺³	M	cACCNon-isolated	Hydrocephalus, cACC	Karyotype analysis	TOP
29	29	24	F	Dysgenesis CCNon-isolated	Ambiguous genitalia, SEC, FGR,dysgenesis CC	Karyotype analysis,CNV-seq	TOP
30	35	23 ⁺³	F	cACCNon-isolated	Absent CSP, VSD, plsv, aortic hypoplasia, dilated lateral ventricles, cACC	Karyotype analysis, CNV-seq, WES	TOP
31	28	19 ⁺⁴	M	cACCNon-isolated	Cerebellar dysgenesis, arachnoid cysts, clubfeet, micrognathia, NF8mm	Karyotype analysis, WES	TOP

(continued on next page)

Table 1 (continued)

NO.	Maternal age	G (weeks)	Sex	Type of ACC	Prenatal US /MRI	Method	Outcome
32	30	24	M	cACCNon-isolated	Posterior fossa abnormalities, cerebellar dysgenesis, cACC	Karyotype analysis, WES	TOP
33	28	26 ⁺²	M	cACCNon-isolated	Absent CSP, vermian hypoplasia, dilated lateral ventricles, cACC	Karyotype analysis,WES	TOP
34	28	26 ⁺²	M	cACCNon-isolated	Absent CSP, dilated lateral ventricles, upward displacement of the third ventricle, cACC	Karyotype analysis, CNV-seq, WES	TOP
35	31	23 ⁺⁶	F	cACCNon-isolated	Dilated lateral ventricles, cerebellar hypoplasia, cACC	CNV-seq, WES	TOP
36	25	23 ⁺⁴	F	cACCNon-isolated	Absent CSP, dilated lateral ventricles, upward displacement of the third ventricle, cACC	CNV-seq, WES	TOP
37	36	25 ⁺³	M	cACCNon-isolated	Dilated lateral ventricles, upward displacement of the third ventricle, left choroidplexuscysts, cACC	CNV-seq, WES	TOP
38	28	26 ⁺⁴	M	cACCNon-isolated	Polyhydramnios, dilated lateral ventricles, cACC	CNV-seq, WES	TOP
39	33	22	F	cACCNon-isolated	Absent CSP, dilated lateral ventricles, upward displacement of the third ventricle, the longitudinal cisterna of the brain widened, cACC	Karyotype analysis, CNV-seq, WES	TOP
40	22	21	M	cACCNon-isolated	Vermian hypoplasia, cACC	Karyotype analysis, CNV-seq, WES	TOP

Abbreviations: F, female; M, male; CC, the corpus callosum; cACC, complete agenesis of the corpus callosum; pACC, partial agenesis of the corpus callosum; FGR, fetal growth restriction; TOF, tetralogy of fallot; CSP, the cavum septi pellucidi; NT, nuchal translucency; EIF, echogenic Intracardiac Focus; VSD, ventricular septal defect; SEC, subependymal cysts; SUA, single umbilical aretery; TOP, termination of pregnancy; LB, livebirth.

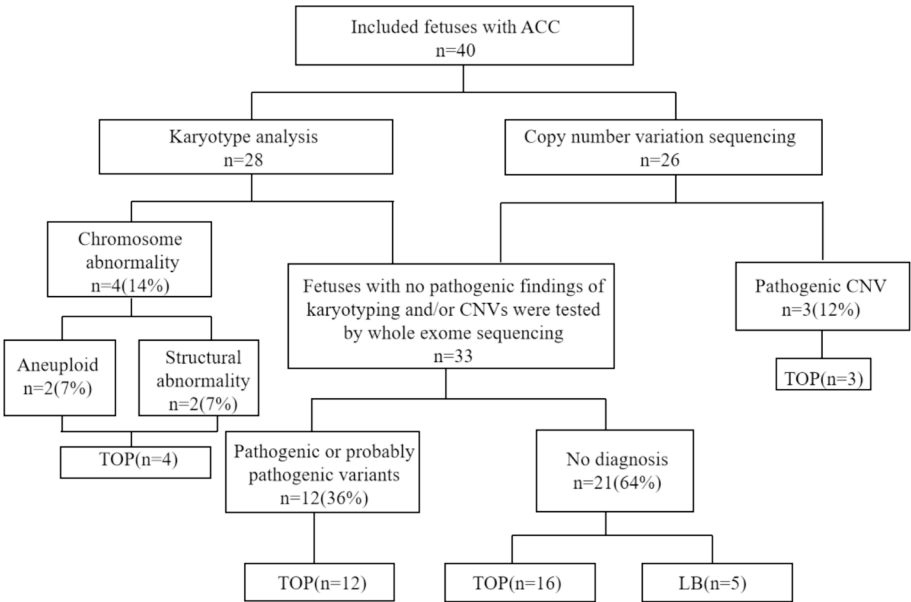


Fig. 1. Flowchart of the study design of the research. ACC: agenesis of the corpus callosum; CNV-seq: copy number variation sequencing; CNV: copy number variation; WES: whole-exome sequencing; TOP: termination of pregnancy; LB: live birth.

obtained during pregnancy. In the fetal ACC diagnosed by imaging during pregnancy, WES results provided critical prognostic information, transforming the fetal prognosis from severe intellectual impairment with high penetrance to a high probability of normal development. These results significantly influenced parental decisions, as the pregnant women and their families who had fetal single-gene variants detected in this study decided to terminate their pregnancies (100 %). In 24 % (5/21) of undiagnosed cases, the parents decide to continue the pregnancy. The parent’s decision to continue the pregnancy with a negative WES result played a role despite being told that neurodevelopmental disorders could not be ruled out. Our findings suggest strategies for prenatal counseling and genetic analysis regarding fetal ACC. When karyotype analysis and CNV-seq results were negative, WES was used to investigate ACC’s monogenetic etiology. Using this strategy, couples are able to

decide whether to end the pregnancy if the underlying etiology could be determined. In the event of fetal ACC, the residual risk of intellectual disability may be <20 % when CNV-seq and WES are negative [28]. When genetic analysis reveals no detectable abnormalities, it is crucial to conduct long-term follow-up on children who were diagnosed prenatally with ACC to provide accurate prenatal counseling. This study has some limitations. First, our study is a single-center study, the included samples must undergo WES after CNV-seq/ karyotyping negative, and the sample size is small. Studies with larger sample sizes could lead to more accurate diagnosis of fetal ACC. Secondly, prenatal findings are phenotypes reported by MRI and ultrasound, which may restrict this study’s ACC classification scheme. The fetuses were diagnosed with isolated ACC. however, it is likely that the disease process changes and that more abnormalities manifest in later

Table 2
Chromosomal disorders and genomic disorders.

Case	Karyotype	Additional imaging finding				
25	47,XY,+18	Ventricular septal defect, polyhydramnios, no gastric vesicles, esophageal atresia, abnormal posture of both hands, absence of right radius, bilateral ulnar dysplasia				
26	47,XX,+18	Ventricular septal defect, polyhydramnios, single umbilical aretery, diaphragmatic hernia, ependymal cyst, absent nasabone, overlapping finger				
27	46, XY, rob (13;14)(q10;q10), +13	Extensive periventricular white matter injury, ventricular septal defect, polyhydramnios				
28	47,XX,+mar[36]/47,XX,t (2;7)(p13;p15), +mar[14]	hydrocephalus				
Case	CNV-seq	Type	Size (Mb)	Inheritance	ACMG	Candidate gene
10	Seq[hg19]17q11.2(29100000–30360000) × 1	deletion	1.26	De novo	pathogenic	<i>B9D1</i>
17	Seq[hg19]17p13.3p12(1–13780000) × 3	duplication	13.78	De novo	pathogenic	<i>YEHA</i> <i>PAFAH1B1</i>
29	Seq[hg19]5q35.3(178039794–180683885) × 1	deletion	2.64	De novo	pathogenic	<i>NSD1</i>

Abbreviations: CC, the corpus callosum.

Table 3
Detection and analysis of gene variants in 12 ACC fetuses.

Case	Gene	Transcript RefSeq	Variant				Novel/ reported	Inheritance mode	Acmg	Diagnoses
			Nucleotide change	Amino acid	Mutation type	Origin				
12	<i>L1CAM</i>	NM_024003	c.807-6G > A		Splice	Maternal	Reported	XLR	LP(PS3 + PM2_S)	L1 syndrome (MIM 308840)
13	<i>L1CAM</i>	NM_024003	c.524-1G > A		Splice	Maternal	Reported	XLR	LP(PVS1 + PM2_S)	
14	<i>L1CAM</i>	NM_024003	c.807-6G > A		Splice	Maternal	Reported	XLR	LP(PS3 + PM2_S)	
15	<i>L1CAM</i>	NM_024003	c.1989C > G	p. Tyr663Ter	Nonsense	Maternal	Reported	XLR	LP(PS3 + PM2_S)	
16	<i>TUBA1A</i>	NM_001270399	c.83A > T	p. His28Leu	Missense	Maternal	Novel	AD	LP(PM1 + PM5 + PP3 + PM2_S)	Lissencephaly type 3 (MIM 611603)
17	<i>TUBB2B</i>	NM_178012	c.824G > A	p. Ser275Asn	Missense	De novo	Novel	AD	LP(PS2 + PM1 + PM2_S + PP2)	Complex cortical dysplasia with other brain malformations 7(CDCBM7) (MIM 610031)
18	<i>SON</i>	NM_032195	c.6127C > T	p. Arg2043Ter	Nonsense	De novo	Novel	AD	P(PVS1 + PS2 + PM2_S)	ZTTK syndrome(MIM 617140)
19	<i>RAB3GAP1</i>	NM_012233	c.899 + 1G > A		Splice	Paternal	Reported	AR	P(PVS1 + PM2_S + PP3)	Warburg micro syndrome 1 (WARBM1) (MIM 600118)
			c.997A > T	p. Lys333Ter	Nonsense	Maternal	Reported		P(PVS1 + PM2_S + PM3)	
20	<i>SPTAN1</i>	NM_001130438.2	c.7210G > A	p. Glu2404Lys	Missense	De novo	Reported	AD	P(PS2 + PM1 + PM2_S + PP2)	Early infantile epileptic encephalopathy-5 (EIEE5)(MIM 613477)
21	<i>AMPD2</i>	NM_004037.7	c.2021G > T	p. Arg674Leu	Missense	Paternal	Reported	AR	LP(PM1 + PM2_S + PM3 + PM5 + PP3)	Pontocerebellar hypoplasia type 9 (PCH9) (MIM 615809)
			c.409G > T	p. Glu137Ter	Nonsense	Maternal	Reported		P(PVS1 + PM2_S + PP3)	
22	<i>ZEB2</i>	NM_014795	c.2481-2488dupCAAAAGTA	p.Ile830fs	Frameshift	Maternal	Novel	AD	LP(PVS1 + PM2_S)	Mowat-Wilson syndrome (MWS) (MIM 235730)
23	<i>ZBTB20</i>	NM_001164342	c.1805G > A	p. Gly620Asp	Missense	De novo	Reported	AD	LP(PM5 + PM1 + PM2_S + PP2 + PS2)	Primrose syndrome (PRIMS) (MIM 259050)

Abbreviations: AD, autosomal dominant; AR, autosomal recessive; XLR, X-linked recessive; P, pathogenic; LP, likely pathogenic; CNV-seq, copy number variation sequencing.

gestations or during the early stages after birth. Recently, WES has improved the diagnostic rates of fetal ACC and confirmed the genetic etiology of monogenic factors. With trio-based WES, genetic heterogeneity and VUS have become a challenge for genetic counseling. However, accidental discovery can help couples prepare for future pregnancies. Although the genetic basis for ACC was identified in 48 % of fetuses, its cause remained unknown in 52 %. Further research is needed to elucidate the genetic and biological basis of CC formation, its complete or partial deletion, and associated brain malformations and clinical findings.

Conclusions

This retrospective study conducted comprehensive genetic examinations of fetuses with ACC at the cytogenetic and molecular genetic levels. Genetic etiology, especially monogenic factors, is crucial in ACC development. Genetic results provide vital prognostic information and shift fetus prognosis, significantly influencing parental decisions. The results of this study provide a new theoretical basis for prenatal diagnosis, prognostic assessment, scientific guidance, and recommendations for future pregnancies.

CRediT authorship contribution statement

Hengqing Sun: Writing – review & editing, Writing – original draft, Data curation. **Ke Li:** Formal analysis, Data curation. **Lu Wang:** Resources, Data curation. **Lijuan Zhao:** Resources, Data curation. **Chenyu Yan:** Resources, Data curation. **Xiangdong Kong:** Methodology, Formal analysis. **Ning Liu:** Writing – review & editing, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We are grateful to the patients and their families who provided information and samples for this study.

Funding

This research was supported by the National Natural Science Foundation of China (81701533, 81672110).

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